C51 Test platform introduction:

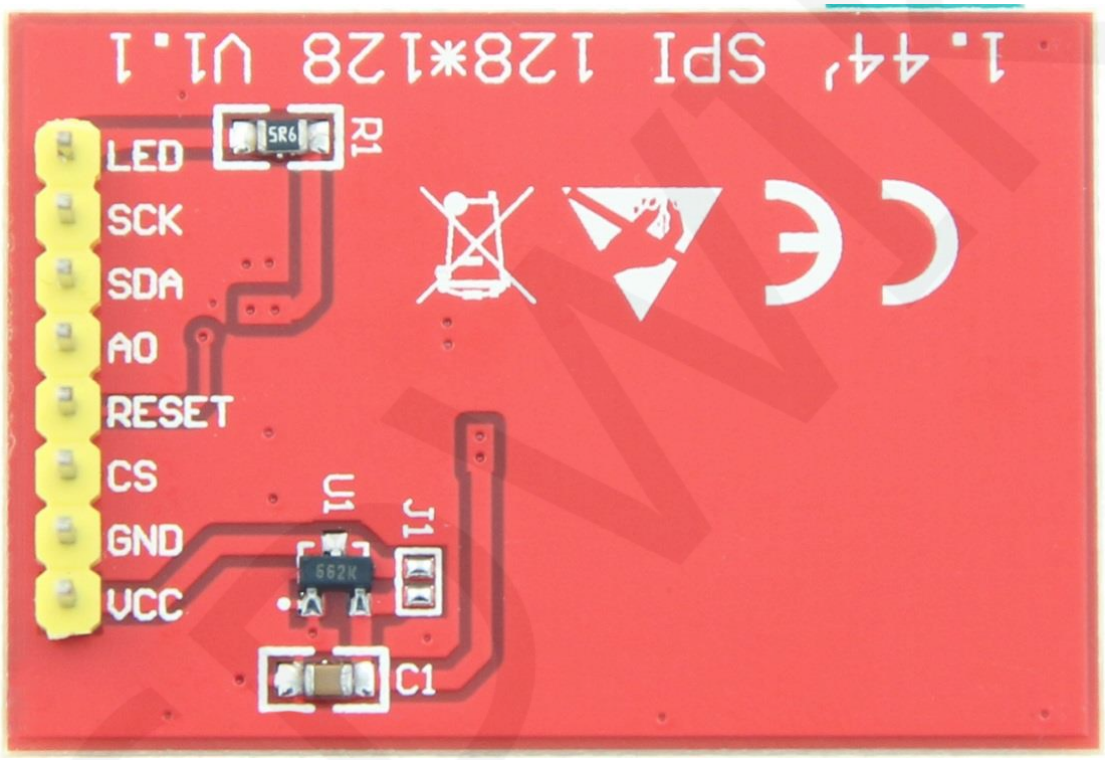
Development board: STC89/STC12 development board

MCU ： STC89C52RC、STC12C5A60S2

Crystal frequency : 12MHZ

Wiring instructions:

The wiring pins of STC89C52RC and STC12C5A60S2 are identical, as follows:



Pin silkscreen picture

STC89C52RC and STC12C5A60S2 microcontroller test program wiring instructions			
Number	**Module Pin**	**Corresponding to STC89/STC12 development board wiring pin**	**Remarks**
1	**VCC**	5V/3.3V	LCD power supply is positive (3.3V~5V)

2	**GND**	GND	LCD Power ground
3	**CS**	P13	LCD selection control signal
4	**RESET**	P33	LCD reset control signal
5	**A0**	P12	LCD register / data selection control signal
6	**SDA**	P15	LCD SPI bus write data signal
7	**SCK**	P17	LCD SPI bus clock signal
8	**LED**	P32	LCD backlight control signal (high level lighting, if you do not need control or use STC89C52RC, please connect 3.3V)

Demo function description:

1. This set of test program procedures is applicable to the STC89C52RC and STC12C5A60S2 platforms;
2. This set of test program uses the software SPI and hardware SPI function of the single-chip platform (STC89C52RC only software SPI function);
3. When using the software SPI function or the hardware SPI function, the wiring pin definition is the same, but the initialization is different;
4. Please follow the above wiring instructions to find the corresponding development board and MCU for wiring;
5. STC89C52RC microcontroller RAM is only 25KB, so only a simple brush test, other test items can not be tested;
6. This set of tests supports display switching in four directions. For details, see the display direction switching instructions.
7. STC12C5A60S2 microcontroller test program contains the following test items:
   A. 3D menu display test;
   B. The main interface displays the test;
   C. simple brush test;
   D. rectangular drawing and filling test;

E. circular drawing and filling test;

F. triangle drawing and filling test;

G. English display test;

H. Chinese display test;

I. picture display test;

J. rotating display test;

8. this module does not have a touch function, please remove the touch screen handwriting test items;

Display direction switching instructions:

Find the macro definition USE_HORIZONTAL in lcd.h as shown below:

```
//////////////////////////////////////用户配置区//////////////////////////////////////
#define USE_HORIZONTAL     0//定义液晶屏顺时针旋转方向  0-0度旋转，1-90度旋转，2-180度旋转，3-270度旋转
////////////////////////////////////////////////////////////////////////////////////
```

USE_HORIZONTAL 0 //0° Rotate

USE_HORIZONTAL 1 //90° Rotate

USE_HORIZONTAL 2 //180° Rotate

USE_HORIZONTAL 3 //270° Rotate